AYMANE AIT DADS

Agentic AI Intern | Agentic AI Systems · LLM Fine-Tuning · Industrial AI Engineering

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, $106K+ prize pool) by designing and building an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer with BF16 precision and 8,192-token chain-of-thought context. Reduced manual network-analysis time by 60% at Orange Maroc by deploying a production-grade Random Forest and K-Means clustering pipeline across 100K+ fiber-optic records, translating model outputs directly into engineering recommendations. Brings hands-on expertise in agentic pipelines, LLM orchestration, RAG, tool-calling, and Claude API integration - exactly the stack TE Connectivity's AI Hub requires to build AI systems for industrial engineering problems. Aims to design and prototype agentic AI systems at the intersection of frontier AI and industrial R&D, contributing to scalable platform tooling used by engineers across TE's global operations.

CORE COMPETENCIES

Agentic AI Systems | LLM Orchestration | Prompt Engineering | RAG Pipelines | LLM Fine-Tuning | Agentic Pipelines | Prototype & Experiment | Industrial AI Engineering

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built an end-to-end agentic pipeline combining Random Forest classification and K-Means clustering on **100K+ fiber-optic network records** to map customer performance profiles to 5 commercial service tiers used directly by the network optimization team.
- Reduced **manual analysis time by ~60%** through automated feature engineering, model evaluation workflows, and reproducible Python scripts replacing analyst-driven review processes.
- Designed a clustering pipeline that translated latency, jitter, and upload-speed signals into **5 actionable service-tier segments**, enabling data-driven maintenance recommendations for thousands of network nodes.
- Delivered **stakeholder presentations** translating model outputs and diagnostic visualizations into concrete engineering recommendations, bridging data science outputs and operational decision-making using Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge *Kaggle Competition · In Progress (Jun 2026) · Ranked 17th / 3,000+ teams*

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) targeting all attention projections, MLP layers, and lm_head.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers to produce 605 verified cipher rows and 64 verified bit-manipulation rows for clean CoT training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection *EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in EURECOM class*

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M parameters) with a custom MLP head, dual learning rates, and selective unfreezing of up to 8 transformer blocks across 250K training images from 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; applied a 3-model ensemble weighted by validation AUC and 10-view TTA (flips, crops, blur, resizes) with FP16 mixed precision, achieving stronger out-of-distribution generalization at 1 epoch than at 4 epochs (0.793 vs. 0.767 AUC).

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya *2021 - 2023*

SKILLS

Agentic AI & LLM Engineering: LangChain | Agentic Pipelines | LLM Orchestration | RAG | Tool-Calling | Claude API | Prompt Engineering

ML Frameworks & Fine-Tuning: PyTorch | Hugging Face Transformers | LoRA / QLoRA | Unsloth | TRL / SFTTrainer | vLLM | PEFT

Data, MLOps & Infrastructure: Python | Pandas | Scikit-learn | Docker | Supabase | PostgreSQL | Git | Power BI